

Measurement of Protein Concentration

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Biochemical research often requires determination of protein concentration in solutions. Several techniques for such measurement have been developed; most of them, however, have limitations due to varied sensitivity or reaction specificity to amino acids in the protein. Since the amino acid content varies from protein to protein, no single assay will be suitable for all protein samples.

A quick, useful estimation of protein concentration is to simply read the Absorbance (OD, optical density) at 280 nm of the protein solution using ultraviolet spectrophotometry. Protein absorb at 280 nm due to the presence of tyrosine and tryptophan residues. Since the content of these two amino acids varies markedly from protein to protein, the extinction coefficient, usually expressed either as $E_{280}^{1\%}$ or $E_{280}^{1\text{ mg/mL}}$ varies considerably.

When protein is placed in an alkaline system containing Cu^{2+} , a colored complex forms between the peptide bonds of the protein and the copper atoms. Bicinchoninic acid (BCA) is a compound that forms a complex with cuprous ion (Cu^+) in an alkaline environment resulting in a stable, highly colored chromophore with an absorbance maximum at 562 nm. This reaction forms the basis of the BCA protein quantitation system.

In Bradford assay, one chemical reagent, Coomassie Brilliant Blue G-250, is added to protein solution to develop a color whose intensity is measured in a spectrophotometer. A "standard protein" of known concentration is also treated with the same reagents and a calibration curve is plotted.

Neither of the methods is perfect because each is dependent on the amino acid content of the protein. Either of them will provide a satisfactory result if the proper experimental conditions are established and/ or a suitable standard protein is chosen. Other important factors in selection among different methods include the sensitivity and accuracy desired, the presence of interfering substance and the time available for assay.

- Use sample 1 as blank in experiment for spectrophotometer zero setting. Standard samples should be run in duplicate (or triplicate).
- 15 mL of standard protein (1.0 mg/mL) and 3 mL unknown protein solution will be used for Spectrophotometric assay and BCA assay in total for each group.
- Due to the same volume examined in each assay, plot the data of Absorbance (OD) directly against the concentration of standard sample.
- Compare the result derived from the different methods and discuss about it.

The Spectrophotometric assay / Absorbance at 280 nm (A_{280}) - Test tube & cuvette

1. Prepare 14 10-mL test tubes and add reagents (standard proteins or unknown proteins, deionized water) as listed in the Table 1. Mix each sample on a vortex mixer.
2. After mixing, use a **1 mL-plastic cuvettes for UV/vis to read A_{280}** .
3. Prepared a calibration curve by plotting mg/mL of standard protein versus absorbance at 280 nm, which is used to estimate the amount of protein in unknown samples. Calculate the protein concentration of unknown sample.
4. Calculate $A_{1\%}^{1\text{cm}} = \quad (280 \text{ nm})$

Table 1. Spectrophotometric assay for protein concentration

Sample No.	1/1'	2/2'	3/3'	4/4'	5/5'	6/6'	7
Standard proteins 1.0 mg/mL (mL)	0	0.4	0.8	1.2	1.6	2.0	-
Unknown protein (mL)	-	-	-	-	-	-	2.0
Deionized water (mL)	2.0	1.6	1.2	0.8	0.4	0	0
mg/mL protein	0	0.2	0.4	0.6	0.8	1.0	
A_{280}							
Av. A_{280}							

The Bicinchoninic Acid (BCA) Assay-96 well plate

1. Prepare BCA-working solution by mixing 4 mL of Reagent A (BCA solution) with 80 μL Reagent B (Cu^{2+}) in a 5-mL test tube.
2. Dilute standard protein (BSA) solution to 0.5 mg/mL. This reagent will be provided at the concentration at 1 mg/mL. Take 500 μL sample to 500 μL PBS using a 1.5-mL test tube for calibration curve data.
3. Make a serial dilution using 1.5-mL test tubes for standard sample preparation according to Table 2. Repeat step2 and step3 for duplicate.

Table 2. Standard sample preparation for BCA assay

Sample No.	1/1'	2/2'	3/3'	4/4'	5/5'	6/6'
Standard proteins 0.5 mg/mL (μL)	0	50	100	150	200	250
PBS (μL)	500	450	400	350	300	250
mg/mL protein	0	0.05	0.1	0.15	0.2	0.25

4. Add each standard sample respectively to 96-well plate as shown in Figure1 B-G in two columns for duplicate.
5. For experimental sample measurement: add 20 μL experimental sample to two sample wells; add 10 μL sample to two sample wells and add 10 μL PBS in the two wells for 1/2 dilution; add 5 μL sample to two sample wells and add 15 μL PBS in the two wells for 1/4 dilution. (Dilution in this step also can be performed in 1.5-mL test tubes beforehand. Prepare with a moderate excess of sample solution.)

- Add 200 μL BCA-working solution to each sample well. Incubate at 37°C for 15-20 min. Read at 562 nm in a microplate reader to measure A_{562} .
- Take notes in Table 2 and plot curve using the values obtained from the standards OD_{562} , calculate the protein concentration of unknown protein solution.

A				
B		1: 20 μL PBS	1': 20 μL PBS	7: 20 μL unknown protein
C		2: 20 μL standard sample 2/2'	2': 20 μL standard sample 2/2'	7': 20 μL unknown protein
D		3: 20 μL standard sample 3/3'	3': 20 μL standard sample 3/3'	8: 10 μL unknown protein+ 10 μL PBS (1/2 dilution)
E		4: 20 μL standard sample 4/4'	4': 20 μL standard sample 4/4'	8': 10 μL unknown protein+ 10 μL PBS (1/2 dilution)
F		5: 20 μL standard sample 5/5'	5': 20 μL standard sample 5/5'	9: 5 μL unknown protein+ 15 μL PBS (1/4 dilution)
G		6: 20 μL standard sample 6/6'	6': 20 μL standard sample 6/6'	9': 5 μL unknown protein+ 15 μL PBS (1/4 dilution)
H				

Figure 1 Layout of 96-well plate using for protein concentration determination

■ step 3 and 4; ■ , ■ , ■ step 5.

Table 2. BCA assay for protein concentration

Sample No.	1/1'	2/2'	3/3'	4/4'	5/5'	6/6'	7/7'	8/8'	9/9'
Standard proteins (μL)	0	20	20	20	20	20	-	-	-
Unknown protein (μL)	-	-	-	-	-	-	20	10	5
PBS (μL)	20	-	-	-	-	-	0	10	15
mg/mL protein	0	0.05	0.10	0.15	0.20	0.25	-	-	-
A_{562}									
Av. A_{562}									

Notes:

When calculate the concentration of unknown protein, it is important to multiply by dilution factor.

*Bradford Assay will be used in Creatine Kinase activity measurement. The protocol list below is only for reference. Please think about the **dyeing mechanism**.*

The Bradford MicroAssay- Test tube & cuvette

The Bradford Assay, based on protein binding of Coomassie Brilliant Blue dye in acidic solution causing a shift in wavelength of maximum absorption ($\lambda_{\max}$) of the dye from 465 nm to 595 nm.

- 1 mL of standard protein (1.0 mg/mL), 1 mL unknown protein solution and 60 mL of Coomassie Brilliant Blue dye and will be used for Bradford Microassay for each group.
1. Dilute standard protein solution to 0.1 mg/mL for about 10 mL.
 2. Prepare 17 test tubes and add reagents (standard proteins or unknown proteins, deionized water and Coomassie Brilliant Blue) as listed in the Table 1. Mix each sample on a vortex mixer.
 3. After addition of dye solution to a protein sample, color development is complete in 2 min and the color remains stable for up to 1 hour. Use a **3 mL-plastic cuvettes (for vis)**, with a wavelength from 380 to 780 nm (visible spectrum) to measure A595.

Note:

If the same cuvette is used for reading samples, it is a good idea to read those samples with less protein content first to reduce error arising from Bradford dye carryover in the cuvette as a result of incomplete rinsing.

All samples should be read sequentially following the color reaction development.

4. Prepared a calibration curve by plotting μg (or mg/mL) of standard protein versus absorbance at 595 nm, which is then used to estimate the amount of protein in unknown samples. Calculate the protein concentration of unknown sample.

Table 3. Bradford assay for protein concentration

Sample No.	1/1'	2/2'	3/3'	4/4'	5/5'	6/6'	7/7'	8	9	10
Standard proteins 0.1 mg/mL (mL)	0	0.1	0.2	0.4	0.6	0.8	1.0	-	-	-
Unknown protein (mL)	-	-	-	-	-	-	-	0.05	0.1	0.2
Deionized water (mL)	1.0	0.9	0.8	0.6	0.4	0.2	0	0.95	0.9	0.8
Bradford working Buffer (an acid solution of Coomassie Brilliant Blue G-250)	3	3	3	3	3	3	3	3	3	3
μg protein	0	10	20	40	60	80	100			
A ₅₉₅										
Av. A ₅₉₅										